



Why do we need this GenC learning Program?

Gen C learning program engages young talents with a comprehensive learning pathway, giving the millennials an opportunity to interact with Subject Matter Experts (SME), understand the corporate environment, and groom themselves.

Cognizant emphasizes on Learner Autonomy where students take charge of their own learning, with the available tools and resources. More focus is on “learning” than “teaching”. Get ready to embark your own learning adventure!

Program at a glance

Learning consisting of 2 Stages:

- Stage 1 – Functional testing & Salesforce/MSCRM
- Stage 2 – Java programming concepts & Selenium Automation Concepts
- Business Aligned Project

Interim Evaluation

Final Evaluation

Program Highlights

- The complete learning journey is formalized using adult learning principles, where problem solving and applying the skills gained are given more importance than conceptual learning.
- Learner Autonomy is encouraged via Flipped Classroom, where the learning platform offers world class learning resources, and students would not be constrained by tutelage of an instructor.
- Get mentored by SME, whose motivation and guidance will help you accelerate in the learning journey.
- This program is applicable to Interns as well as GEN Cs.

Learning Journey with Flipped Classroom

This program encourages you to be more autonomous learners during guided self-learning hours, completing the learning objectives on your own pace and style, and get ready for the hands-on practice time.

The complete learning path is set in the [GEN C Learn Platform](#), which you can login with SSO.

Flipped Classroom

Self-Learning Time

- Go through the Learning Objectives
- Try to accomplish the learning objectives by accessing learning resources

Practice Time

- Get guidance from Subject Matter Expert
- Deep dive on to the learning concepts and solve a problem statement

Recommended Program Sequence

The learning journey contains 2 stages, followed by a Business Aligned Project.

- Stage 1 – Functional testing & Agile Fundamentals & Salesforce
- Stage 2–Java programming Fundamentals & Selenium Automation Concepts.
- Business Aligned Project will provide you an experience of real time problem solving in Agile methodology.

Stage 1 – Functional testing & Salesforce/MSCRM

Day 1 to 5

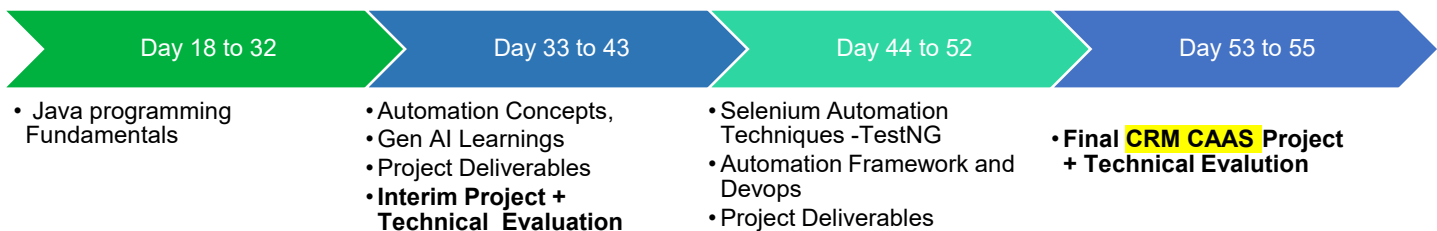
- Functional testing & Agile Fundamentals

Day 6 to 15

- Salesforce/MSCRM
- ICT assessment

Day 16 & Day 17

- **Qualifier Assessment**



Key Learning Components of the Program

Cognizant has collaborated with Udemy to provide world class learning videos for the evolving future of work. These Udemy programs are woven into a learning path, empowering you to plan and learn at your style.

The program also connects you with Subject Matter Experts (SMEs) to get the professional guidance on your queries in the learning journey.

The program doesn't ONLY concentrate on the technical skilling, but also on the shaping up of the Behavioral skills. **Behavioral learning** would be done in ILT mode, with few Self-paced learning modules too.

Evaluation Model

The program continuously evaluates if you are able to apply those self-learnt skills to solve a real-time business problem. Depicted below are the four key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

- Interim Evaluation (Project + Technical) through Video Interview
- Final Evaluation (Project + Technical) through Video Interview

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Code Challenges and ICTs will help you to enhance your expertise level.

The weightage of the Interim Evaluation will be 30%, while the Final Evaluation will account for 70%. However, the performance status of GenC will be indicated using the RAG (Red Amber Green) system. In this system, 'Green' signifies meeting the passing criteria, which is equal to or greater than 70%, the benchmark score for all evaluations.

Interim & Final Evaluation Approach

Below is the Evaluation Structure for GenC Learning Journey

The interim evaluation will be held halfway through the learning journey, while the final evaluation will take place at the end of the learning journey.

During the interim evaluation, the GenC will be interviewed by a Technical Subject Matter Expert (SME) from the Business Unit (BU) to assess your knowledge through a technical discussion. Additionally, the Integrated and Mini project completed by the GenC will also be evaluated by the BU SME. Please note that the Mini project is an individual activity and not a group activity.

During the final evaluation, the GenC will again be interviewed by a Technical SME from the BU to assess their knowledge through a technical discussion. In the same evaluation call, the integrated and **CRM Salesforce case study** project completed by each group will be evaluated by the BU SME. Please note that the Hackathon project is a group activity and not an individual activity.

Preparatory Learning

For this automation track training, here are some preparatory learning materials that can be helpful for the interim and final technical evaluations. Preparatory learning material refers to resources, materials, and content that you can use to prepare. These materials are designed to provide you with the necessary knowledge, skills, and understanding to succeed in your learning objectives.

Remember, apart from these learning materials, actively participating in the training sessions, asking question and practicing hands-on exercises will greatly contribute to your overall learning and preparedness for the evaluations.

Software Installation Instruction:

Refer to the provided Selenium automation environment installation setup document for step-by-step instructions on configuring and setting up the necessary tools and dependencies.

Program Completion Criteria

Stage 1 (Pre-requisites)
Gating Criteria: Qualifier Assessment

Stage 2 & Beyond (Advanced Skills)
Gating Criteria: Performance Health Status is Green

GenC/Intern Training	Evaluation Components	Pass Criteria	Evaluation Done by
Performance Health Status - PHS (only from Stage 2)	Interim Evaluation (Project + Technical)	Green, 1 Attempt	BU SME
	Final Evaluation (Project + Technical)	Green, 2 Attempts*	

Outcome of Interim / Final evaluation will be RED, AMBER or GREEN status.

Note: 100% Completion of Hands On in Stage 1 is mandatory for qualifier assessment and 100% Completion of Hands On in Stage 2 is mandatory for interim / final evaluation eligibility.

Key Check Point Intervals in the Learning Journey

Progressing to Stage 2 depends on clearing the qualifier assessment after stage1. Candidates who do not clear the Stage 1 Qualifier will be terminated from the Internship. However, based on the demand and later needs, they will be considered for the CSD mode of training.

Subsequent stages learning journey, your progress will be measured. On the below check point intervals, your overall Performance Health Score will be calculated as on date, and the RAG status will be arrived.

Table - Check Point Intervals

Check points	Interpreting Status
Interim Evaluation	- Green - On Track for Graduation - Red /Amber - There will not be any re-attempts given
Final Evaluation	- Green - On Track for Graduation - Red /Amber – Only 1 attempt are given (Re-attempt is not applicable if the student is Red in Interim and Final Evaluation) Note: If student fails after the applicable re-attempts, they will be considered as “Not Graduated”.

Icebreaker



Icebreaker session will be conducted for a duration of initial **5 days**. During the session, various topics related to Corporate Induction, Talent Management, Cognizant Agenda on

Core Values, Leader Talks, Alumni, BU Mentor connects will be covered. Followed by icebreaker, technical training will kick start.

Following sessions will be covered during the **5 days** of icebreaker

- Corporate Induction
- Talent Manager Connect
- Cognizant Agenda Sessions on Core Values
- Leader Talks (Academy) and many more...

A recommended day-wise schedule is provided below for the learning, with the learning content for the day, the practice hands-on and extended hands-on to be done for the day or any other activities are listed. Few days might be interleaved to accommodate the extension due to Behavioral Training.

GenAI-Program Overview

Introduction

AI Accelerate is a comprehensive program designed to empower learners with the knowledge and skills needed to harness the transformative potential of Artificial Intelligence (AI). This handbook serves as a guide, providing essential information and resources to help learners navigate through the program successfully. From understanding the fundamentals of AI to apply the advanced concepts in real-world scenarios. AI Accelerate is your gateway to unlock the power of AI and shaping the future of technology.

Program overview.

AI Accelerate offers a learning opportunity, allowing GenCs to engage learning through the self-paced learning, Expert connect, and Knowledge assessment to measure the skills.

Focus areas

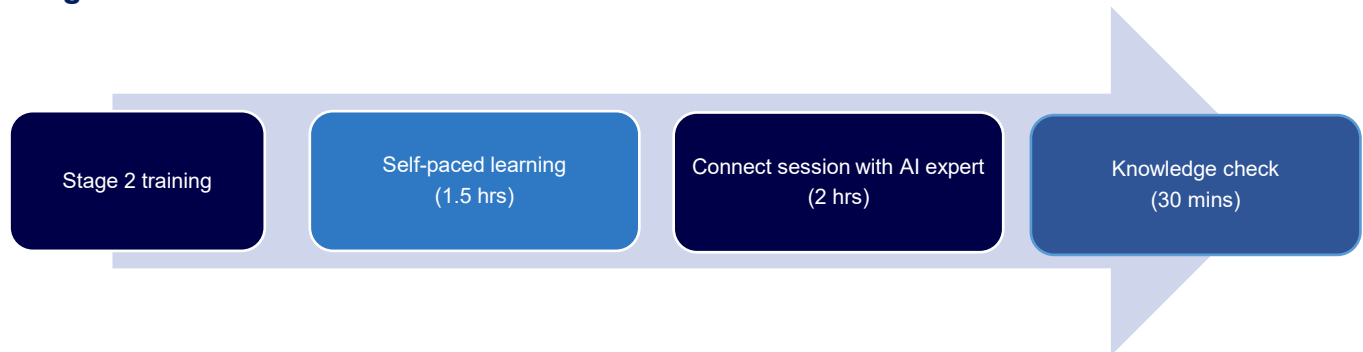
- **Learning:** GenCs will have access to curated content and resources that cover a wide range of topics related to AI, including best practices, and case studies. This learning aspect aims to deepen GenCs understanding of AI and its applications in various industries.
- **Expert connect:** GenCs will have the opportunity to connect with expert in the field of AI. This expert will provide guidance, support, and insights to help GenCs navigate their learning journey and gain valuable insights into the industry.
- **Practice sessions:** GenCs can practice the use cases provided sessions that are designed to reinforce their learning and help them apply their knowledge in real-world scenarios. These sessions will provide GenCs with hands-on experience and practical skills that are essential for success in the field of AI.

Performance outcomes

Upon completing the self-paced learning component of AI Accelerate GenCs are expected to achieve the following performance outcomes:

- GenCs will demonstrate a thorough understanding of the fundamentals of AI, including key concepts, terminology, and principles.
- GenCs will be able to apply AI concepts and techniques to solve real-world problems, demonstrating their ability to analyze, design, and implement AI solutions.
- GenCs will develop and apply critical thinking and problem-solving skills in the context of AI, enabling them to identify, analyze, and address complex challenges.
- GenCs will collaborate effectively with peers, mentors, and industry experts to achieve common goals and contribute to the advancement of AI knowledge and practice.

Program workflow



Schedule – Stage 1: Week 1

Week 1 will be focusing on Functional testing & Agile Fundamentals.

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Day 1 to 5

Agile Functional Testing

Topic #	Topic Name	Topic Objective	References
Module 1	Agile Software Testing Concepts	Understanding SDLC and STLC	Theory and Foundations
		Distinguishing between Project and Product in testing context	
		Exploring V-Model and the roles of QA, QC, and QE	
		Differentiating between White Box and Black Box Testing	
		Delving into Static Testing and Dynamic Testing	
		Conceptualizing Verification and Validation	
		System Testing Types (GUI Testing)	
		System Testing Types (Functional vs. Non-Functional Testing)	

		Test Design Techniques and strategies	Includes Re-Testing, Regression Testing, Exploratory Testing, Adhoc, Sanity, Smoke Testing, and End-To-End Testing.
		Understanding STLC phases (Use case, Test scenario, Test case)	
		Understanding STLC phases (Test Environment, Execution, Defect Reporting)	
		Understanding STLC phases (Test Closure, and Test Metrics)	
Module 2	Agile Software Testing Project	Introduction to the Live Project	Practical Application
		Extracting Functional Requirements from FRS	Requirement Analysis
		Creating comprehensive Test Scenarios and Test Cases	Writing Test Artifacts
		Execution of Test Cases	
		Reporting and Tracking Bugs	Defect Management
		Test Sign-off	Completion Phase
Module 3	Agile/Scrum Process	Overview of Agile and Scrum methodologies	Agile Fundamentals
		Understanding Scrum Teams, Sprints, and User Stories	
		Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'	
		Exploring Sprint Activities (Planning, Refinement, Review, Retrospective)	
Module 4	JIRA Tool Implementation	Installation and Configuration of JIRA	Hands-on Tool Setup
		Project creation in Jira and user management	
		Creating Backlog, Epic, and User Stories in JIRA	Requirements in JIRA
		Sprint creation and management	
		Understanding the Sprint life cycle in JIRA	
		Bug creation in Jira	Defect Reporting in JIRA
		Writing test cases in JIRA	Test Case Management
		Creating Test Cycles and executing/updating Test Cases in Jira	Test Execution in JIRA

Continuous Learning: Technical Enablement

Learn the basics of Agile Fundamentals and Software Testing Life Cycle



[Learn Manual Software Testing + Agile with Jira Tool](#)

Learn all the sections in this Udemy course to gain deep understanding about below modules

Module : Software Testing Concepts

- Understanding Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC)
- Distinguishing between Project and Product in the context of testing
- Exploring V-Model and the roles of QA (Quality Assurance), QC (Quality Control), and QE (Quality Engineering)
- Differentiating between White Box and Black Box Testing
- Delving into Static Testing and Dynamic Testing
- Conceptualizing Verification and Validation in software testing
- System Testing Types, including GUI Testing and Functional vs. Non-Functional Testing
- Test Design Techniques and strategies like Re-Testing, Regression Testing, Exploratory Testing, Adhoc Testing, Sanity, Smoke Testing, and End-To-End Testing
- Understanding Software Testing Life Cycle (STLC) phases, including Use case, Test scenario, Test case, Test Environment, Execution, Defect Reporting, Test Closure, and Test Metrics.

Module : Software Testing Project

- Introduction to project
- Extracting Functional Requirements from Functional Requirement Specifications (FRS)
- Creating comprehensive Test Scenarios and Test Cases
- Execution of Test Cases
- Reporting and Tracking Bugs
- Test Sign-off

Phase : Agile Testing + Jira Tool

Agile/Scrum Process:

- Overview of Agile and Scrum methodologies
- Understanding Scrum Teams, Sprints, and User Stories
- Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'
- Exploring Sprint Activities such as Sprint Planning, Backlog Refinement, Sprint Review, and Sprint Retrospective

Jira Tool:

- Installation and Configuration of JIRA
- Project creation in Jira and user management
- Creating Backlog, Epic, and User Stories in JIRA
- Sprint creation and management
- Understanding the Sprint life cycle in JIRA
- Bug creation in Jira
- Writing test cases in JIRA with Zephyr plugin
- Creating Test Cycles and executing/updating Test Cases in Jira

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- The Importance of QEA and Career Scope
 - The Scope Debate: Quality Assurance vs. Software Development
 - Trainer-Guided Integrated Project: Activity Readiness
 - Trainer-Guided Integrated Project: Configured Jira Project & Backlog Management
 - Trainer-Guided Integrated Project: Test Case Development in Jira (using Zephyr/similar plugin)
 - Trainer-Guided Integrated Project: Test Execution & Status Reporting Artifacts
 - Trainer-Guided Integrated Project: User story document
 - Trainer-Guided Integrated Project: Retrospection Report
 - Trainer-Guided Integrated Project: Sprint Backlog
 - Placeholder to submit other Functional agile deliverables
- ✓ **You should have introduced more than one sprints**
 - ✓ **Ensure you have in-sprint automation concepts introduced as part of project**
 - ✓ **Ensure you have created Regression suite and smoke suite**
 - ✓ **Ensure your regression suite consists of minimum 25 test cases and test scripts**
 - ✓ **Ensure your smoke suite covers end to end business functionality**

Additional Learning:

Technical Quizzes: Functional Testing

Day 5

Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Functional Testing** skill appear for the **Functional Testing -ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform. Attempts allowed to practice the assessment is 3.

Assessment Duration: 4 hrs

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

Schedule – Stage 1: Week 2

Week 2, 3 will be focusing on Salesforce/MS CRM

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Day 6

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement

Learn the basics of Salesforce.



[Salesforce 101: Introduction to Salesforce](#)

Refer the section in this Udemy course and complete the corresponding learnings.

- Introduction
- Salesforce Functional Capability
- The Clouds
- Career in Salesforce

Learn & Practice

- Data Management & Chatter concepts
- Process Automation

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- BU Driven Handson

Day 7,8

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement



[Salesforce Certified Administrator Training \(Jan 2025\)](#)

Refer the section in this Udemy course and complete the corresponding learnings.

- Introduction
- Configuration & Setup
- Object Manager and Lightning App Builder – My First App
- Workflow/Process Automation
- Salesforce Security
- Analytics Management
- Data Management

- Salesforce Marketing Applications
- Service & support Applications
- Productivity & Collaboration

Learn & Practice

- Sales Cloud
- Reports & Dashboard
- Salesforce Security

Continuous Learning: Technical Hands-on Mandatory Hands-on

- BU Driven Handson

Day 9,10

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement



[Salesforce Certified Administrator Training \(Jan 2025\)](#)

Refer the section in this Udemmy course and complete the corresponding learnings.

- Introduction
- Configuration & Setup
- Object Manager and Lightning App Builder – My First App
- Workflow/Process Automation
- Salesforce Security
- Analytics Management
- Data Management
- Salesforce Marketing Applications
- Service & support Applications
- Productivity & Collaboration

Learn & Practice

- Service Cloud
- Intro to Salesforce CRM & Org Setup with User Mgmt

Continuous Learning: Technical Hands-on Mandatory Hands-on

- BU Driven Handson

Schedule – Stage 1: Week 3

Day 11,12

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement



[Salesforce Certified Administrator Training \(Jan 2025\)](#)

Refer the section in this Udemy course and complete the corresponding learnings.

- Introduction
- Configuration & Setup
- Object Manager and Lightning App Builder – My First App
- Workflow/Process Automation
- Salesforce Security
- Analytics Management
- Data Management
- Salesforce Marketing Applications
- Service & support Applications
- Productivity & Collaboration

Learn & Practice

- Salesforce Health Cloud Overview

Continuous Learning: Technical Hands-on Mandatory Hands-on

- BU Driven Handson

Day 13

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement



[Salesforce Certified Administrator Training \(Jan 2025\)](#)

Refer the section in this Udemy course and complete the corresponding learnings.

- Introduction

- Configuration & Setup
- Object Manager and Lightning App Builder – My First App
- Workflow/Process Automation
- Salesforce Security
- Analytics Management
- Data Management
- Salesforce Marketing Applications
- Service & support Applications
- Productivity & Collaboration

Learn & Practice

- Veeva CRM
- Salesforce Health Cloud for Provider
- nCino Training

Continuous Learning: Technical Hands-on Mandatory Hands-on

- BU Driven Handson

Day 14

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement



[Salesforce Certified Administrator Training \(Jan 2025\)](#)

Refer the section in this Udemy course and complete the corresponding learnings.

- Introduction
- Configuration & Setup
- Object Manager and Lightning App Builder – My First App
- Workflow/Process Automation
- Salesforce Security
- Analytics Management
- Data Management
- Salesforce Marketing Applications
- Service & support Applications
- Productivity & Collaboration

Once completing Udemy course, Learnings, all Handson & Practice assessments related to Salesforce/MSCRM skill appear for the **Mock Qualifier Assessment**. This will help you to learn the pattern of the actual assessment in the platform.

Assessment Duration: 15 minutes
Assessment Attempts: 3
No.of Questions appear in each attempt : 5

Day 15

Salesforce/MSCRM (Day6 -Day15)

Continuous Learning: Technical Enablement



[Salesforce Certified Administrator Training \(Jan 2025\)](#)

Refer the section in this Udemy course and complete the corresponding learnings.

- Introduction
- Configuration & Setup
- Object Manager and Lightning App Builder – My First App
- Workflow/Process Automation
- Salesforce Security
- Analytics Management
- Data Management
- Salesforce Marketing Applications
- Service & support Applications
- Productivity & Collaboration

Continuous Learning: Technical Hands-on Mandatory Hands-on

- BU Driven Handson

Schedule – Stage 1: Week 4

Day 16 & Day 17

Once completing Udemy course, Learnings, all Handson & Practice assessments related to Salesforce/MSCRM skill appear for the following below Assessment.

Qualifier Assessment on '(SFDC Testing - Knowledge Based Assessment [101-BASICS] - AEST649_NP)'

Assessment Duration: 60 Minutes
No.of MCQ Questions: 40

Schedule – Stage 2: Week 4

Day 18 to 27 will be focusing on Core Java .

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Continuous Learning: Technical Hands-on through Tekstac

Post Completion of the ILT session, Trainee must Practice the Handson provided in the Tekstac Platform . In build code editor is available in this platform and Handson are automatically evaluated when the code is compiled in the editor.

Continuous Learning: Java Practice Hands-on in Eclipse

You will Parallel Learn and complete the Tekstac Platform Hands-on exercises and practice Topic wise Java Practice Handson Daily . This Exercise is set for practice only and not mandate . But completion of Tekstac Platform Handson is mandate for enablement of Assessments. Follow these steps to practice - Topic Wise Java Practice Hands-on :

1. Software: Download the Eclipse software.
2. Practice Topic-wise Hands-on Exercises: Practice the given topic-wise hands-on exercises in your local environment using Eclipse daily post completion of Learning and platform driven Handson Exercise.
3. Submit for Evaluation: Once you have completed the exercises, submit the solutions in .zip file in platform for evaluation.

Day 18

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic
Chapter-1	Introduction to Java, Data Types and Variables	History & Features of Java: OOP, platform independence, JVM, JDK, JRE Data Types: Primitive and non-primitive types, Variable declaration, Type casting
Chapter-2	OOP Basics	Object-Oriented Principles: Encapsulation, Inheritance, Polymorphism, Abstraction Classes and Objects: Class structure, Object creation, Instance variables Constructors: Default constructor, Parameterized constructor, Constructor overloading

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Streamline Your Health Journey
- Water Tank Capacity
- Square Series
- Brilliant Restaurant

Continuous Learning: Java Practice Hands-on in Eclipse

Basic Syntax

- Print an integer value
- Command line arguments
- Print the input from scanner

Looping Constructors

- Find the factorial of a given number
- See the output as 0, 1, 1, 2, 3

- Print the below output
- Print the triangle of numbers

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

Section 2: Java Programming

Day 19

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic
Chapter-3	Inheritance, Abstraction & Encapsulation	Inheritance: Single inheritance, Multilevel inheritance, super keyword
		Polymorphism: Method overloading, Method overriding
		Abstraction: Abstract classes, Interfaces
		Encapsulation: Access modifiers (public, private, protected), Encapsulating data

Continuous Learning: Technical Enablement through Self-learning

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Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Investment Calculation
- Dazzle Closet
- Lego Builders
- Plan Smart

Continuous Learning: Java Practice Hands-on in Eclipse

Function and Array

- Factorial of a number using recursion
- Find the biggest number among 1,2,3,4,5,65,76,5,4,33,4,34,232,3,2323

Object Oriented Programming

- Object Oriented Programming

- Constructors
- Abstract Classes
- Interfaces
- Static Members
- Final Keyword

Basic Arithmetic Operations

- Java program to swap two numbers
- Convert from Fahrenheit to Celsius
- Swap two numbers without using a third variable
- Java program to add two numbers
- Java program to find the GCD of two numbers
- Java program to find the LCM of two numbers
- Java program to print the sum of digits of a number

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

Section 2: Java Programming

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic
Chapter-4	Keywords	Static Members: Static variables, Static methods, Static blocks
		this and super: Referring current object, Accessing parent class methods
		final Keyword: Final variables, Final methods, Final classes
Chapter-5	Arrays,Strings,StringBuffer & StringBuilder	Arrays: One-dimensional array, Multidimensional array, Array traversal and operations
		Strings: String class, Immutability, String methods: length, substring, equals, etc.
		StringBuffer & StringBuilder: Mutability, Common methods: append, insert, delete

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Mythic Rhythms
- WordZee
- Fishing competition

Continuous Learning: Java Practice Hands-on in Eclipse

Control Flow Statements

- Demonstrate if...else statement
- Demonstrate nested if...else if... statement
- Demonstrate nested if...else statement
- Find odd and even numbers
- Printing the prime numbers
- Check whether a given number is Armstrong
- Check if a number is prime
- Check if a number is a palindrome
- Check if a number is an Armstrong number

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

Section 2: Java Programming

Day 20

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic
Chapter-5	Arrays,Strings,StringBuffer & StringBuilder	StringBuffer & StringBuilder: Mutability, Common methods: append, insert, delete
Chapter-6	Exception Handling	Types of Exceptions: Checked exceptions, Unchecked exceptions try-catch Block: Single try-catch, Multiple catch, finally block throw and throws: Manually throwing exceptions, Propagating exceptions

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Mass of Rocket Component
- Fitness Tracker
- Array Aggregate
- Product Details

Continuous Learning: Java Practice Hands-on in Eclipse

Loops

- Demonstrate for loop
- Print stars using for loop, where the number of stars printed should be equal to the row number
- Demonstrate while loop
- Print the entered number in reverse
- Demonstrate the usage of break statement inside while loop
- Print the alphabets using for loop
- Demonstrate the usage of break and continue statements inside while loop
- Demonstrate for each loop
- Print Floyd's triangle
- Print Pascal's triangle
- Print the Fibonacci series
- Printing the Multiplication table
- Find the factorial of a number
- Print the prime numbers up to a given number
- Calculate the power of a number using a while loop
- Verify whether a number is a perfect number or not
- Printing the Fibonacci series from 1 to 10
- Find the greatest of three numbers
- Print the largest two numbers from an array of given numbers

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

Section 2: Java Programming

Additional Learning:

Technical Quizzes:

- Java Operator, Control flow statement

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter-6	Exception Handling	Custom Exceptions: Creating custom exceptions
Chapter-7	Java Collections	Introduction to Collections: Collection hierarchy, Collection interfaces (List, Set, Map)
		List Interface: ArrayList, LinkedList, Vector, Stack

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- College Namelist
- Babitha's App
- Alta Motors

Continuous Learning: Java Practice Hands-on in Eclipse

Method Overloading and Overriding

- Final methods be overloaded
- Static methods be overloaded
- Final methods be overridden
- Static methods be overridden
- Overload a main method in Java
- Object u have created

POLYMORPHISM & INHERITANCE

- Demonstrate method overloading
- Demonstrate method overriding
- Demonstrate inheritance
- Demonstrate the use of the 'super' keyword

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 2 in this Udemy course and complete the corresponding learnings.

Section 2: Java Programming

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter-7	Java Collections	Set Interface: HashSet, LinkedHashSet, TreeSet
		Map Interface: HashMap, LinkedHashMap, TreeMap, Hashtable
		Iterator: Iterator, ListIterator

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Java Practice Hands-on in Eclipse

Arrays

- Find the largest number
- Remove duplicates from an array
- Find the second largest number in an array?
- Sort an array using bubble sort
- Sort an array using selection sort
- Sort an array using insertion sort
- Implement binary search
- Implement linear search
- Find the sum of elements in an array
- Find the average of elements in an array
- Find the maximum and minimum elements in an array
- Find duplicate elements in a Java Array
- Find the smallest and largest numbers in a Java Array
- Java program to swap two strings without using temp or third variable
- Demonstrate creating an array
- Duplicate elements in an array of numbers
- Demonstrate creating a multidimensional array
- Print the elements of the array in reverse
- Print alternative elements in a String array
- Find the greatest number in an integer array
- Find the least number in an integer array
- Using a for loop with single-dimensional arrays

- Using a for-each loop with single-dimensional arrays

Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Additional Learning:

Serial and Parallel Sorts in Java

- <https://www.geeksforgeeks.org/serial-sort-vs-parallel-sort-java/>

Streams and Optionals

- <https://www.geeksforgeeks.org/java-8-stream-tutorial/?ref=lbp>

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter-7	Java Collections	Comparable & Comparator: Sorting with Comparable, Sorting with Comparator
Chapter-8	Java Thread Handling	Introduction to Threads: Thread vs Process, Thread lifecycle
		Creating Threads: Extending Thread class, Implementing Runnable interface
		Thread Methods: sleep(), join(), start(), run()
Chapter-9	Java File Handling	Character Streams: FileReader, FileWriter, BufferedReader, BufferedWriter

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Copy Names
- Booking System

Practice Section – Integrated Capability Test

Once completing Udemy course, Learnings, Hands-on related to Java skill appear for the Practice Assessment. This assessment has been set up in platform to help you to assess and apply the concepts learnt as part of Java skill.

You can take up this assessment as a practice till your qualifier assessment for practice. The

purpose of this practice assessment is to prepare you for the actual assessment and enhance your information retention. Regular practice helps reinforce what you've learned, making it easier to recall during the actual assessment.

Assessment Attempts: Unlimited

Assessment Duration: 3 hrs

No.of Questions appear in each attempt : 3

Continuous Learning: Java Practice Hands-on in Eclipse

String Handling

- Java program to compare two strings
- Java program to reverse a string
- Java program to count the number of vowels and consonants in a string
- Java program to find the frequency of characters in a string
- Java program to check if two strings are anagrams
- A string such that the first character is uppercase, the second is lowercase, and so on the numbers from a string and print only the alphabets
- Reverse the words in a sentence
- The frequency of words in a paragraph

Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Day 22 - 25

Core Java

Continuous Learning: Technical Enablement

Chapter	Chapter Name	Topic
Chapter-9	Java File Handling	File Handling: File class, File reading and writing (FileInputStream, FileOutputStream)
Chapter-10	JDBC (Java Database Connectivity)	JDBC API Overview: JDBC architecture, Driver Manager, Statement and PreparedStatement
		CRUD Operations: Insert, Update, Delete, Select Queries
		Transactions: Commit, Rollback, Savepoints

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Animalia
- Customer Records

Continuous Learning: Java Practice Hands-on in Eclipse

Exception Handling

- Demonstrate exception handling using try-catch
- Demonstrate the use of finally block
- Demonstrate the use of throw and throws
- Demonstrate custom exceptions
- Demonstrate the advantage of finally in Exception Handling
- Custom class which is immutable

Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter-11	Java 8 Features	Lambda Expressions: Lambda syntax, Functional interfaces
		Stream API: Stream operations (filter, map, reduce)
		Optional Class: Avoiding NullPointerException with Optional
Chapter-12	Generics & Annotations	Generics: Generic classes, methods & wildcards
		Annotations: Built-in annotations (@Override, @Deprecated) & Custom annotations

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Fit Freak
- Employee Details
- Student Score Info
- Vintage Books Emporium

Continuous Learning: Java Practice Hands-on in Eclipse

Collections Framework

- Java program to demonstrate the use of collections (ArrayList)
- Java program to demonstrate the use of collections (HashMap)
- Java program to demonstrate the use of collections (HashSet)
- Java program to demonstrate the use of collections (LinkedList)
- Java program to demonstrate the use of collections (TreeSet)
- Java program to demonstrate the use of collections (PriorityQueue)
- Java program to demonstrate the use of collections (Stack)
- Java program to demonstrate the use of collections (Vector)

Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Day 26 - 28

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter-12	Generics & Annotations	Annotations: Built-in annotations (@Override, @Deprecated) & Custom annotations
Chapter-13	Design Patterns	Design Patterns: Singleton, Factory, Observer, Strategy patterns

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Mystic Line
- Area And Volume

Continuous Learning: Java Practice Hands-on in Eclipse

Collections Framework

- Java program to demonstrate the use of collections (Deque)
- Java program to demonstrate the use of collections (LinkedHashMap)
- Java program to demonstrate the use of collections (LinkedHashSet)
- Find duplicates in a list
- Two HashMaps for equality
- Creating an ArrayList
- Creating a LinkedList
- Creating a HashSet
- Creating a LinkedHashSet
- Creating a TreeSet
- Creating a PriorityQueue
- Creating a HashMap using the Map interface
- Creating a LinkedHashMap
- Creating a TreeMap

Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Continuous Learning: Technical Hands-on

Additional Learning:

Technical Quizzes:

- Applying Object Oriented Concepts in java

Core Java

Continuous Learning: Technical Enablement through Tekstac

Mandatory Learning

Java 8 Features - Streams and Optionals. Asynchronous and Parallel Programming in Java 8.
Go through web pages for learning below specific topics

[Asynchronous and Parallel Programming](#)

[Java User Input \(Scanner\)](#)

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- The Perfect Password
- Library Management

Additional Learning



[Java In-Depth: Become a Complete Java Engineer!](#)

Go through only below mentioned sections 16 & 17 and implement the examples along with the author

- Section 16: Collections Framework (aka Data Structures)
- Section 17: Generics

JDBC

Continuous Learning: Technical Enablement Learn and Practice



[Java Database Connection: JDBC and MySQL](#)

- Go through entire course.
- Implement the examples along with the author.

Additional Learning: Technical Quizzes:

- Collections Framework
- Advanced Java Concepts

Note : On Day 26 - Once completing Udemey course, Learnings, all Handson & Practice assessments related to **Java** skill appear for the **Java -Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assess-Type-1: Code Challenge – Java - Group 1

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt: 1

Note : On Day 29 –

Assess-Type-1: Code Challenge – Java - Group 2

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt: 1

Day 29 & 30

Core Java

Continuous Learning: Technical Enablement

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

[Java Database Connection: JDBC and MySQL](#)

- Rehearse the sections based on need

Day 31 & 32

Core Java

Continuous Learning: Technical Enablement

Learn and Practice



[Learn Selenium with Java, Cucumber & Frameworks](#)

[Java Database Connection: JDBC and MySQL](#)

- Rehearse the sections based on need

Note : On Day 31 - Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Java** skill appear for the **Java - ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

GenC - QAQE-TECHNICAL-JAVA TRACK - CORE JAVA SKILL WAR - SKILL BASED ASSESSMENT [101-BASICS] -LAB_ATKJE072

Assessment Duration: 4 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 3

GenAI (Day 32)

Continuous Learning: Technical Enablement

AI Accelerate

- By this course you will complete GenAI self-paced learning , AI expert connect session and AI Knowledge check assessment.

Program completion criteria

Everyone must register for the following e-learning course on [C-Learn](#) and complete a KBA assessment on Moodle to successfully finish this program.

Online learning: C-Learn



Activity Code: **ELRNG01863**

Fundamentals of Generative AI [101-Basics]

This AI course is designed to equip learners with:

- The foundational knowledge and skills required to harness the power of Generative AI.
- The ability to identify opportunities for innovation and implementation of AI within their organizations.
- The skills to drive organizations toward a future of enhanced creativity and competitive advantage using AI techniques.

Schedule – Stage 2: Day 33 to 55

Day 33 to 35 will be focusing on Selenium configuration and WebDriver Basics.

Udemy learnings are recommended in the Platform to understand the fundamental concepts.

Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Business Aligned Project (Mini Project/ Hackathon)

As the selenium learning starts, the project details (Mini project and the Hackathon) will be given for the learners so that they can parallel keep doing the project activities along with the rest of the learnings.

- Mini project in selenium is a great way to deepen your understanding, showcase your skills, and prepare for real-world automation tasks.
- It allows to apply theoretical knowledge of Selenium and web automation in a practical, real-world scenario. Working on a project helps you gain hands-on experience with Selenium WebDriver, locators, and other essential components.
- Building a mini project helps you understand the design and implementation of automation frameworks, such as Page Object Model (POM) and Data-Driven Testing.
- You can integrate Selenium with other tools like TestNG for test management and reporting.

Selenium Automation for Realtime Project				
Section #	Module / Topic	Detailed Lecture / Sub-Topic	Concept/Tool	Notes
1	Setting the Project Foundation	Automation Project Setup and Environment Configuration	Java, Maven/Gradle	Dependencies and Build Automation.
		Introduction to Test Design Patterns	Page Object Model (POM)	Structure, benefits, and implementation.
		Setting up Logging and Reporting Tools	Log4j, Extent Reports	Generating detailed, professional reports.
2	Advanced WebDriver Concepts	Synchronization Techniques	Explicit & Fluent Waits	Handling dynamic elements and improving stability.
		Handling Complex Web Elements	Action Class, JavaScript Executor	Mouse and Keyboard interactions, scrolling.
		Handling Popups, Windows, and Frames	WebDriver	Switching contexts and managing alerts.
		Capturing and Analyzing Failure Points	Taking and Organizing Screenshots	Capturing evidence upon test failure.
3	Designing the Automation Framework	Hybrid Framework Design	POM + Data-Driven + Keyword-Driven	Combining best practices into one framework.
		Data Management for Live Projects	Data-Driven Testing	Reading test data from Excel, JSON, or databases.
		Implementing Robust Exception Handling	Java	Writing code that handles unexpected errors gracefully.
		Writing Reusable Utility Methods	Custom Libraries	Creating helper functions for common tasks (e.g., file handling).
4	Test Execution & Reporting	Parallel Test Execution	TestNG/Selenium Grid	Running multiple tests simultaneously to save time.
		Grouping and Prioritizing Tests	TestNG Annotations	Running Smoke, Sanity, or Regression Suites.
		Finalizing and Interpreting Execution Reports	Extent/Allure Reports	Analyzing results for pass/fail metrics.
5	Real-Time DevOps Integration	Integrating Framework with Version Control	Git/GitHub	Pushing and managing the project code.
		Scheduling and Running Tests in CI	Jenkins	Setting up jobs for automated nightly/daily runs.

Mini Project & Hackathon – Objective

- **CI/CD Integration:** Integrating your Selenium tests with CI/CD tools like Jenkins helps automate the testing process as part of the build pipeline.
- **Automated Testing:** Ensures that your application is tested automatically with every code change, improving software quality.
- **Reporting and Logging:** Generate detailed test reports that provide insights into test execution, pass/fail status, and error details.
- **Logging:** Implement logging to capture detailed information about test execution, which is useful for debugging and analysis.
- **Reusable Code:** Writing reusable and maintainable code is a key aspect of any project, and a mini project helps you practice this.
- **Collaboration:** It encourages collaboration and teamwork, as you can divide tasks among team members and integrate their work using Selenium.
- **Exception Handling:** You'll learn to handle exceptions and errors that occur during test execution, making your scripts more robust.
- **Debugging Skills:** Debugging skills are enhanced as you troubleshoot issues in your automation scripts etc.

The given Mini Project and the Hackathon are expected to be completed on or before the end week of the program.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Note:

Mini Project is an individual team member activity. Project Requirement is available in the

platform. Candidates must complete the requirement and submit the solution for evaluations by BU SME.

Hackathon is an individual activity. Project requirement is available in the platform. Candidates must complete the requirement and submit the solution for evaluations by BU SME.

Day 33

Automation Concepts, configuration, WebDriver Basics

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 1	Selenium Basics and overview	Introduction to Browser Automation
		What is Selenium and Overview
		History and evolution of Selenium
		Selenium Components (WebDriver vs Selenium IDE vs Selenium Grid)
		Supported programming languages (Java, Python, C#, etc.)
		Cross-browser and cross-platform compatibility.
		Advantages and limitations of Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Refer to and learn the same topics covered by the trainer on Udemy.
2. Practice the Tekstac Platform Handson.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find Element by Id
- Find Element by Name

Learn the basics of WebDriver Basics



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Day 34

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 2	Selenium Architecture	What is Selenium Webdriver: 1) WebDriver is one of the component in selenium. 2) WebDriver is a java interface. 3) WebDriver is an API(Application Programming interface)
		Understanding the WebDriver Architecture / API methods. WebDriver, RemoteWebDriver, ChromeDriver , FirefoxDriver, EdgeDriver etc..
Chapter 3	Setting up Selenium environment	Installing Java Development Kit (JDK) and Eclipse IDE
		Downloading Selenium WebDriver
		Configuring Selenium WebDriver with Eclipse

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning)hands-on exercises provided in the local environment.
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 2_Handson
- Chapter 2_PPT Learning Ref.Content
- Chapter 2_Video Learning Ref.Content
- Chapter 3_Handson
- Chapter 3_PPT Learning Ref.Content
- Chapter 3_Video Learning Ref.Content



Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: Technical Enablement

Chapter	Chapter Name	Topic
Chapter 4	Selenium WebDriver	Locating GUI Elements in Application Under Test (AUT)
		How to identify the locators in web page
		First Selenium Webdriver Script
		Launching Browsers in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment Submit the solutions for evaluation.
3. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find options in dropdown
- Find Radio Element

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 4_Handson

- Chapter 4_PPT Learning Ref.Content
- Chapter 4_Video Learning Ref.Content



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

Day 35

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 5	Locating various Web elements	Find Element by Id,Name,Link Text,Partial Link Text,Class Name,CSS (Tag , ID , Class , Attribute , etc..)
		Handling Shadow DOM in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find Element by Tag Name
- Handling Check Box
- Find Element by Class Name
- LinkText and PartialLinkText

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 5_Handson
- Chapter 5_PPT Learning Ref.Content
- Chapter 5_Video Learning Ref.Content[Part-1]
- Chapter 5_Video Learning Ref.Content[Part-2]
- Chapter 5_Video Learning Ref.Content[Part-3]



Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25

Section 3: Selenium WebDriver

Additional Learning:

Technical Quizzes:

Quiz 1 - Automation Concepts, Selenium configuration, WebDriver Basics

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 5	Locating various Web elements	Relative Locators and Chained Locators in Selenium
		Locating By Xpath (Single Slash, Double Slash , contains(), starts_with() , text() , Last())

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy
- Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure
- Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios)
- Trainer-Guided Integrated Project: Test Execution Reports and Status
- Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Practice the Assessment.
5. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Learn the basics of WebDriver Basics



Refer section 3 in this Udemy course and complete the corresponding Subtopics from 17 to 25.

Section 3: Selenium WebDriver

Day 36

Selenium Automation Techniques, Dynamic XPath

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 5	Locating various Web elements	Find Element by Id,Name,Link Text,Partial Link Text,Class Name,CSS (Tag , ID , Class , Attribute , etc..)
		Handling Shadow DOM in Selenium
		Relative Locators and Chained Locators in Selenium
		Locating By Xpath (Single Slash, Double Slash , contains(), starts_with() , text() , Last())
		Find options in dropdown. Handle dropdown without using Select Class
		Find Radio Element
		Find Element by Tag Name
Chapter 6	Desired capabilities	Desired Capabilities in Selenium WebDriver

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- CSSLocator
- AbsoluteXpathLocator
- XPathAncestor

- Working with Alerts
- Work with Javascript Executor

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 6_Handson
- Chapter 6_PPT Learning Ref.Content
- Chapter 6_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

Note : On Day 36 - Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium-Web Driver** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge

Assess-Type-1:Code Challenge - Automation Concepts, Selenium configuration, WebDriver Basics

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

Selenium Automation Techniques, Dynamic XPath

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 7	selenium webdriver commands and Web elements interactions	Selenium WebDriver- Commands
		Text fields and buttons
		Find Element VS FindElements in Selenium WebDriver - check order
		Handling Canvas Elements in Selenium
		Using JavaScript Executor in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Handling RegEx Selenium
- Drag and Drop with Slider and Actions and wait
- Selenium – ScreenShot
- Selenium - DatePicker(Actions Class)
- FindElements

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 7_Handson
- Chapter 7_PPT Learning Ref.Content
- Chapter 7_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

Selenium Automation Techniques, Dynamic XPath

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 8	WebDriver Actions	Mouse and keyboard actions
		Drag and drop
		Handling frames and windows
		mouse hover (tool tip) in Selenium
		Handling tabs & cookies

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Handling ToolTip
- Working with different XPath conditions
- PurchaseApp iframe
- Multiple Window
- Selenium - Implicit Wait
- Selenium - Fluent Wait

Continuous Learning: Technical Hands-on

Best Practices- Real Time Technical Hands-on

- Desired Capabilities
- Shadow_DOM_Scenario1
- Shadow_DOM_Scenario2
- Relative Locators and Chained Locators in Selenium

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 8_Handson
- Chapter 8_PPT Learning Ref.Content
- Chapter 8_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

Project – Mini Project Case Study (Day 36 – 42)

Overall Duration: Should start your mini project parallel with the Stage 2 Selenium Automation learnings and practice daily.

The outcomes of doing **Mini Project** are:

- Enables learners to know on the environment setup
- Any web application is taken, and learner try to automate given scenario using Selenium APIs
- Exhibits learner skills on automation of real time applications for smaller requirement.

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Enhanced Automation Test Scripts
- Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting
- Trainer-Guided Integrated Project: Utility Class for Advanced Functions
- Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts

Day 37 & 38

Selenium Automation Techniques, Dynamic XPath

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 9	Multiple windows & frames	Switching between windows
		Switching to frames & iFrames
	WebDriver Waits	Implicit waits
		Explicit waits
		Fluent waits

Chapter	Chapter Name	Topic
Chapter 10	Selenium Automation Techniques	Checkbox Handling and Form Submission
		Locating Web Element Attributes using CSS Selector
		Locating Web Element Attributes using Absolute XPath Locator
		Identifying Ancestors of Web Elements
		Working with alerts
		Working with Java script executor
		Working with Regular expression
		Working with Drag and Drop and Slider
Chapter 11	Advanced WebDriver techniques	Selenium - ScreenShot
		Handling JavaScript/window Alerts
		Headless browsers and drivers in Selenium
		Handling Dynamic Elements in Selenium
		1) get methods , 2) conditional methods, 3) browser methods, 4) navigational methods, 5) wait methods
		How to Find All/Broken links using Selenium Webdriver

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Take Up the Assessment.
5. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 9_Handson
- Chapter 9_PPT Learning Ref.Content
- Chapter 9_Video Learning Ref.Content
- Chapter 10_Handson
- Chapter 10_PPT Learning Ref.Content
- Chapter 10_Video Learning Ref.Content
- Chapter 11_Handson
- Chapter 11_PPT Learning Ref.Content
- Chapter 11_Video Learning Ref.Content

Continuous Learning: Technical Hands-on through Tekstac

Best Practices- Real Time Technical Hands-on

- Relative Locators and Chained Locators in Selenium_Scenario 1
- Relative Locators and Chained Locators in Selenium_Scenario 2
- Relative Locators and Chained Locators in Selenium_Scenario 3
- Relative Locators and Chained Locators in Selenium_Scenario 4



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 3 in this Udemy course and complete the corresponding Subtopics from 26 to 32.

Section 3: Selenium WebDriver

Project – Mini Project Case Study

- Integrate the mini project.

Additional Learning:

Technical Quizzes:

Quiz 1 - Selenium Automation Techniques, Dynamic Xpath

Day 39 – 42

You will be focusing on Selenium WebDriver with POM and ApachePOI

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Selenium Webdriver With POM and ApachePOI

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 12	Calendar using action	Selenium - DatePicker(Actions Class)
	Apache POI	Apache POI - Read Operation : Excel Read Operation
		Apache POI - Write Operation: Excel Write Operation
		Applying POI POM- Flight reservation

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Excel Read Operation
- Excel Write Operation
- Applying POI - Flight reservation

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 12_Handson
- Chapter 12_PPT Learning Ref.Content
- Chapter 12_Video Learning Ref.Content

Learn the basics of WebDriver with POM and ApachePOI



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 4 in this Udemy course and complete the corresponding learnings.

Section 4: Apache POI - Data Driven Testing using MS Excel

Selenium Webdriver With POM and ApachePOI

Learn the basics of WebDriver with POM and ApachePOI



Refer section 4 in this Udemy course and complete the corresponding learnings.

Section 4: Apache POI - Data Driven Testing using MS Excel

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- POM with Page Factory Model

Continuous Learning: Technical Hands-on

Best Practices- Real Time Technical Hands-on

- Mouse_Keyboard_Actions_Cookie_Scenario 1
- Mouse_Keyboard_Actions_Cookie_Scenario 2
- Mouse_Keyboard_Actions_Cookie_Scenario 3
- Mouse_Keyboard_Actions_Cookie_Scenario 4
- Mouse_Keyboard_Actions_Cookie_Scenario 5
- Mouse_Keyboard_Actions_Cookie_Scenario 6
- Mouse_Keyboard_Actions_Cookie_Scenario 7

Selenium Webdriver With POM and ApachePOI

Continuous Learning: Technical Enablement

Learn the basics of WebDriver with POM and ApachePOI



Refer section 4 in this Udemy course and complete the corresponding learnings.

Section 4: Apache POI - Data Driven Testing using MS Excel

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation)
- Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI)
- Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts
- Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation)

Stage : GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

- Fundamentals of Generative AI [101-Basics]
- Generative AI Quick Assessment for Elearning Quiz [101-Basics]
- Introduction to Responsible AI
- Responsible AI-Deep Dive [101- BASICS]

- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.

- Prompt Engineering Foundation - PY0009562

- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

Introduction to GitHub Copilot

 [GitHub Copilot for Beginners: AI Coding Crash Course](#)

Go through entire course. Implement the examples along with the author.

 [GitHub Copilot - The Complete Guide – 2025](#)

Go through entire course. Implement the examples along with the author.

- **Usecase and Handbook & Comprehensive GenAI Hands-On for Automation Developers:** These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

Project – Mini Project Case Study

- Integrate the mini project.

Note : On Day 39 - Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Selenium – Xpath** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge

Assess-Type-1: Code Challenge - Selenium Automation Techniques, Dynamic Xpath

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

Note : On Day 42 - Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Selenium – Apache POI** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge

Assess-Type-1: Code Challenge - Selenium WebDriver with POM and Apache POI

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

MOCK Evaluation through AI BOT (Day 42)

This is an auto-scheduled, AI-driven mock interview, intended to be taken before the interim evaluation. You will receive an auto-scheduled email notification three hours prior to the evaluation.

Details:

- Platform: Mail-driven platform
- Scope: AI will ask questions based on everything you have learned up to date
- Duration: 1.5 hours
- Questions: 15 skill-based Coding questions

Once you complete the evaluation, AI will provide feedback on your performance. This helps you to appear for Interim Evaluation.

Project – Mini Project Case Study

- Integrate the mini project.

Deliver your project and get the review comments from SME.

Day 43 & 44

Evaluation:

Interim Project Evaluation + Technical Evaluation

Automation Testing - Selenium with TestNG

You will be focusing on Selenium with TestNG

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 13	TestNG Concepts	TestNG Assertions
		Introduction to TestNG
		Creating TestNG test cases
		TestNG annotations and assertions
		Data-Driven Testing in TestNG
		Running Tests in Parallel using TestNG

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- TestNG Assertions
- TestNG with two test classes
- Contact Book- POM structure
- TestNG Annotation With Priority
- TestNG Dependency

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 13_Handson
- Chapter 13_PPT Learning Ref.Content
- Chapter 13_Video Learning Ref.Content

Learn the basics of Selenium with TestNG



Refer section 5,6,7 in this Udemy course and complete the corresponding learnings.

Section 5: TestNG Framework for Selenium

Section 6: Page Object Model

Section 7: Hybrid Driven Framework Development with Project (eCommerce Application)

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Integrated & Structured Test Suite using TestNG Framework
 - ✓ You should have automated more than one sprints
 - ✓ Ensure you have in-sprint automation concepts introduced as part of project
 - ✓ Ensure you have created Regression suite and smoke suite automation
 - ✓ Ensure your regression suite consists of minimum 25 test cases and test scripts
 - ✓ Ensure your smoke suite covers end to end business functionality
- Trainer-Guided Integrated Project: Robust Test Data Management via TestNG DataProviders
- Trainer-Guided Integrated Project: Comprehensive Execution Control via testng.xml
- Trainer-Guided Integrated Project: Advanced Reporting with Extent Reports & Listeners
- Trainer-Guided Integrated Project: Implementation of Assertions (Hard and Soft)

Project - Best Practices- Mandatory Real Time Technical Project (Day 44-46)

- Integrate the Real Time Technical Project.

Automation Testing - Selenium with TestNG

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 13	TestNG Concepts	Multi-browser testing using TestNG*
		Rerun failed tests*
		Set Test priority in TestNG
		Dependency in TestNG
		Soft /Hard Assertion in TestNG
		TestNG Report Generation in Selenium WebDriver*
		Customize, PDF & Email TestNG Reports in Selenium WebDriver*
	TestNG Class	TestNG with two test classes
	Data driven	Data driven from XML + POM : Contact Book- POM structure

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Cargo Shipping
- Shopify Registration
- XMLParsing -1
- XMLParsing -2
- XMLParsing -3

Learn the basics of Selenium with Datasource XML Parsing (With XML & JSON)



[Learn API Technical Writing: JSON and XML for Writers](#)

Refer sections 1, 2, 3 and 4 in this Udemy course and complete the corresponding learnings.

Section 1: Introduction

Section 2: JSON

Section 3: XML

Section 4: Final Words

Rehearse the below Lend a hand enablement given in platform

- XMLParser_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- XML Parsing

Automation Testing - Selenium with TestNG

Continuous Learning: Technical Enablement

Learn the basics of Selenium with Datasource JSON Parsing (With XML & JSON)

Rehearse the below Lend a hand enablement given in platform

- JSONParsing_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- JSON Parsing
- Json Solution Explanation

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- JSONParsing – 1
- JSONParsing – 2
- JSONParsing – 3

NOTE: On Day 48, Once completing Udemy course, Learnings, all Hands-on & Practice assessments related to **Selenium – TestNG** skill appear for the **Code Challenge Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge - Assess-Type-1:Code Challenge - Selenium with TestNG

Assessment Duration: 2 hrs.

Assessment Attempts: 3

No. of Questions appear in each attempt : 1

Project Deliverable (Hackathon Project) (Day 47 – 52)

Overall Duration: Should start Hackathon project end of Automation Framework milestone and continue doing till Digital Technologies.

The outcomes of doing **Hackathon** are:

- Explore the Innovative Implementations.
- Implement Best practices such as creating **Smoke and Regression suite**.
- Implement **Maven** on the created automation test scripts
- Explore **Jenkins** to execute the test scripts periodically on Selenium Grid.
- Explore Selenium **Grid** to run test scripts on different platforms and against different browsers.

Integrate Jenkins with version controller (**GIT**) and scheduled builds to run automatically.

Day 48 to 52

Automation Framework

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 14	Test Automation Frameworks	Introduction to Automation Frameworks
		Data-Driven Testing in Selenium
		Keyword-Driven Testing Framework in Selenium
		Page Factory
		Building a Modular Testing Framework in Selenium
		Hybrid Framework
	Design Patterns (POM)	Understanding Page Object Model (POM) in Selenium
		Creating Page Classes in Selenium
		Implementing Page Factory in Selenium
		Design Patterns in Automation using Selenium (Singleton, Factory etc.)
Chapter 15	Test Data Handling	Reading and Writing Data to Files (e.g., Excel, CSV , XML, JSON) using Java
		Test Data Parameterization in Selenium
Chapter 16	Test Report and Test Log	Generating Detailed Test Reports (Extent Reports, Allure Reports)
		Screenshot Capture Strategies in Selenium
		Custom Logging with Log4j in Selenium

CAAS Final project Case studies

Before your final evaluation interview, you should have completed below case studies

- CAAS_Classic_Account Case Study
- CAAS_Classic_E2E Case Flow for Case Study
- CAAS_Classic_Sales_LeadCreation Case Study
- CAAS_Lightening_E2E_Service Case Creation
- CAAS_Lightening_Service Campaign Creation

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 14_Handson
- Chapter 14_PPT Learning Ref.Content
- Chapter 14_Video Learning Ref.Content[Part-1]
- Chapter 14_Video Learning Ref.Content[Part-2]
- Chapter 14_Video Learning Ref.Content[Part-3]

- Chapter 15_Handson
- Chapter 15_PPT Learning Ref.Content
- Chapter 15_Video Learning Ref.Content

- Chapter 16_Handson
- Chapter 16_PPT Learning Ref.Content
- Chapter 16_Video Learning Ref.Content[Part-1]
- Chapter 16_Video Learning Ref.Content[Part-2]
- Chapter 16_Video Learning Ref.Content[Part-3]

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer section 7 in this Udemy course and complete the corresponding Subtopics

Section 7: Hybrid Driven Framework Development with Project (eCommerce Application)

Mandatory Hands-on

- Trainer-Guided Integrated Project: The "Sprint Automation" Hybrid Framework Codebase
- Trainer-Guided Integrated Project: Complete Automated Regression/Sprint Suite
- Trainer-Guided Integrated Project: Enterprise-Level Reporting Suite (Extent & Allure)
- Trainer-Guided Integrated Project: CI/CD Ready Execution Artifacts
- Trainer-Guided Integrated Project: Finalized Agile Documentation & Handover Package

Devops

You will be focusing on Devops (Maven , Git, Jenkins, Grid).

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Click Here to View the [BU Mandatory Technical Learning and Handson](#) Document.

Table of Contents:

Topic #	Topic Name	Topic Objective	References
Behavior-Driven Development (BDD)	Cucumber	Introduction to BDD and Gherkin Syntax	Writing feature files and step definitions.
	Cucumber	Integrating BDD with Selenium Framework	Running scenarios and hooks.
Build Automation & Dependencies	Maven	Introduction to Maven	Understanding Project Object Model (POM).
	Maven	Managing Dependencies and Plugins	Setting up Selenium and other libraries.
	Maven	Executing Tests using Maven Goals	Running tests from the command line.
Version Control System	GIT	Introduction to GIT and GitHub	Basic commands (clone, add, commit, push, pull).
	GIT	Branching and Merging Strategies	Collaborative development practices.
	GitHub	Integrating Test Code with Remote Repository	Pushing the automation framework to the cloud.
Continuous Integration/Delivery	Jenkins	Setting up Jenkins CI Server	Installation and configuration.
	Jenkins	Creating a Jenkins Pipeline	Integrating GIT and Maven projects.

	Jenkins	Scheduling and Triggering Automation Runs	Generating test reports in CI environment.
Distributed Test Execution	Selenium GRID	Understanding Selenium Grid Architecture	Hub and Node concepts.
	Selenium GRID	Setting up and Configuring the Grid	Running tests in parallel on multiple machines/browsers.
Containerization	Docker	Introduction to Docker	Understanding images and containers.
	Docker	Running Selenium Tests using Docker Containers	Setting up Dockerized Selenium Grid for consistent environments.
	Docker	Creating a Custom Automation Image	Packaging the complete framework.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 17	Selenium Grid	Introduction to Selenium Grid
		Setting up Selenium Grid
		Running tests on Selenium Grid
		Achieving Parallel Execution in the Cloud
		Setting up selenium grid in a docker/K8S container
	Distributed Execution	Running Tests on Cloud-Based Selenium Grids (e.g., Sauce Labs, Browser Stack)
		Achieving Parallel Execution in the Cloud
	Maven	Introduction to Maven
		Creating and Managing Maven Projects
		Maven Build Lifecycle (e.g., clean, compile, test, package)
	Git Version Control	Introduction to Version Control
		Git Fundamentals (Commits, Branches, Merges)
		Git Workflow (Pull Requests)
		Managing Code Repositories on GitHub or GitLab
	Jenkins CI Systems	Introduction to CI/CD and DevOps
		Jenkins Overview and Jobs Configuration
		Integrating Selenium Tests into Jenkins Pipelines
		Groovy Script Introduction and Syntax

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Hands-on and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Git Config
- Clone Repo
- Add, Commit and Push
- Pull And Merge
- Merge - Resolve Conflict
- Git Tags
- Build Web Application
- Compiling Executing Java using POM

Additional Hands-on

- GIT installation & commands implementation
- Install Jenkins and Creation of new job

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 17_Handson
- Chapter 17_PPT Learning Ref.Content
- Chapter 17_Video Learning Ref.Content[Part-1]
- Chapter 17_Video Learning Ref.Content[Part-2]
- Chapter 17_Video Learning Ref.Content[Part-3]
- Chapter 17_Video Learning Ref.Content[Part-4]
- Chapter 17_Video Learning Ref.Content[Part-5]

Learn Devops and Grid



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections 8 and 10 in this Udemy course and complete the corresponding learnings.

Section 8: CI - Maven, Git, Github & Jenkins

Section 10: Bonus Tips

Section 13: Docker - Bonus Tips

Devops

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment.
3. Submit the solutions for evaluation.

4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Junit With Maven
- Maven Directory Structure
- Maven Shade Plugin

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 18_Handson
- Chapter 18_PPT Learning Ref.Content
- Chapter 19_Handson
- Chapter 19_PPT Learning Ref.Content

Additional Hands-on

- Mavenization using Eclipse M2E plugin Options and CLI command

Learn Devops and Grid



[Learn Selenium with Java, Cucumber & Frameworks](#)

Refer sections 8 and 10 in this Udemy course and complete the corresponding learnings.

Section 8: CI - Maven, Git, Github & Jenkins

Section 10: Bonus Tips

Section 13: Docker - Bonus Tips

Devops

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Udemy.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Enablement

Learn Devops and Grid



Rehearse the section based on need

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- Maven App to access External Service
- Capital Service
- Build project based on another project
- Build Maven Project (Using Freestyle Project)
- Pipeline - compile and test
- Setup Grid and parallel execution

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 20_Handson
- Chapter 21_Handson
- Chapter 21_PPT Learning Ref.Content
- Chapter 22_Handson
- Chapter 23_Handson
- Chapter 24_Handson

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

- Trainer-Guided Integrated Project: Scalable Execution Infrastructure (Selenium Grid & Docker)
- Trainer-Guided Integrated Project: Behavior-Driven Development (BDD) Framework Integration
- Trainer-Guided Integrated Project: Continuous Integration & Delivery (CI/CD) Pipeline Integration

NOTE: On Day 49 - Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium** skill appear for the **MOCK ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Integrated Capability Test (ICT) - SELENIUM MOCK

Assessment Duration: 2 hrs.

Assessment Attempts: 1

No.of Questions appear in each attempt : 1

SELENIUM Assess-Type-2: Integrated Capability Test (ICT)

NOTE: On Day 51, Once completing Udemy course, Learnings, all Handson & Practice assessments related to **Selenium** skill appear for the **Selenium ICT Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assessment Duration: 4 hrs.

Assessment Attempts: 3

No.of Questions appear in each attempt : 1

MOCK Evaluation through AI BOT (Day 52)

This is an auto-scheduled, AI-driven mock interview, intended to be taken before the Final evaluation. You will receive an auto-scheduled email notification three hours prior to the evaluation.

Details:

- Platform: Mail-driven platform
- Scope: AI will ask questions based on everything you have learned up to date
- Duration: 1.5 hours
- Questions: 15 skill-based Coding questions

Once you complete the evaluation, AI will provide feedback on your performance. This helps you to appear for Final Evaluation.

Project – Hackathon (Day 52)

Deliver your project and get the review comments from SME.

The deliverables of the Hackathon will be evaluated by the BU SME.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Schedule – Stage 2: Week 11

Day 53 - 55

Evaluation:

Final CRM CASE Study and Integrated Project Evaluation + Final Technical Evaluation

What is Final Evaluation?

The Final Evaluation will be conducted to certify whether a GenC is eligible to enter into the BU or not. The skill of a GenC will be gauged on the application development and overall technical knowhow towards the end of GenC Training.

Tech SME from BU will be conducting the final tech evaluation. As a fallback, the project mentor can also steer this activity.

The final evaluation will be conducted as two phases. They are the following

1. **Final Technical Evaluation**
2. **Final Project Evaluation**

The mode of these evaluations will be any one of the following:

- F2F(face to face)
- Video Based

1. Final Technical Evaluation (FTE)

The BU Mentor will interview the GenC on various skills achieved throughout the training program and put a score which will be considered for the final PHS of the GenC.

2. Final Project Evaluation (FPE)

In this evaluation, the BU Mentor will be verifying the skills of a GenC on a project perspective. End of this evaluation, the BU Mentor will score the GenC's work based on various evaluation criterions.



How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Udemy courses and by completing the hands-on exercises mentioned in the daily plan.

The below strategies will help you decide the learning approach.

Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

Profile #1



Harry Reacher

Engineering Discipline: Electronics

Skills: Python, Ruby on Rails, nginx

Project: Mining Crime Data to get Route Cause Insights

Learning Approach to Programming Languages: I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

What will work for me?

- Directly complete hands on exercises
- Refer Internet or Udemy Courses
- If hands on are implemented early, clarify your friends questions and troubleshoot their issues

Profile #2



Olivia Richards

Engineering Discipline: Computer Science

Skills: Java, C, C++

Project: Library Management System

Learning Approach to Programming Languages: I have interest, but I don't know where to start.

What will work for me?

- Go through the recommended Udemy Course
- Try completing the hands on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

Profile #3



Greg Anderson

Engineering Discipline: Civil

Skills: C

Project: Fiber reinforced concrete

Learning Approach to Programming Languages: I am scared of programming languages. I haven't got my hands dirty with coding

What will work for me?

- Go through the recommended Udemy Course
- Implement the coding along with the author of the Udemy Course
- Try completing the hands on exercises

- Clarify queries with SME
- Troubleshoot programming issues with help from SME or learner from your classroom whom had already completed

FAQ

1. Who can participate in this program?

Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GEN Cs can participate in this program.

2. Is there any pre-learning I should do?

No. This program is open to all students from any academic discipline.

3. How will I know my RAG status?

It will be shown to you in the GEN C learn Platform, in your Home Page.

4. Whom do I reach out in case of any queries?

Coach is your point of contact.

5. What is the significance of Hands-on in the overall learning journey?

Hands-on focuses on specific topics in a Skill, which you can try and execute in the Platform. Group of such Hands-on exercises will be packaged together as a Code Challenge. This Code Challenge will allow you to benchmark your skills in the learning journey. Hands On/ Code Challenges/ ICT are learning components which will help you in understanding the skills better.

6. What is Code Challenge?

A problem statement will be provided to you and you need to solve it using a single skill.

7. What is Integrated Capability Test (ICT)?

A case study problem statement will be provided to you, that you may need solve using the combination of Skills learnt in the given stage.

8. How many attempts are provided for the Coding challenge and ICTs? Is it open all the time for practice?

The Coding challenges and ICTs are open and there are 3 attempts to take them up.

9. What is the entry criteria for qualifier?
A minimum of 70% hands-on completion and attempt in the CC & ICT is the eligibility criteria for qualifier.
10. What skills are covered in the qualifier?
The skills of Stage 1 are covered in the qualifier. Only ONE attempt is provided to clear with a minimum score of 70%
11. What if I fail in the Interim evaluation?
Your coach will notify your performance in the Interim evaluation. However you can continue with the learning.
12. How many chances will I get in the Final evaluation?
You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.
13. Whom do I reach out in case of any queries?
Coach is your point of contact.